tb_adder_suber.v

source code

```
`timescale 1ns / 1ps

module tb_adder_subtracter();

    reg r_add_nsub;
    reg [3:0] r_a;
    reg [3:0] r_b;
    wire w_carry_out;
    wire [3:0] w_sum;

    adder_subtracter u_adder_subtracter(
        .add_nsub(r_add_nsub),
        .a(r_a),
        .b(r_b),
        .carry_out(w_carry_out),
        .sum(w_sum)
    );

    initial begin

    r_add_nsub = 1'b0; r_a = 4'b0000; r_b = 4'b0000;
    $monitor("time = %0t | mode = %b | a = %d b = %d | carry_out = %b sum
= %b", $time, r_add_nsub, r_a, r_b, w_carry_out, w_sum);


    // 덧셈
    r_add_nsub = 1;
    r_a = 4'd0;  r_b = 4'd0;  #10; // carry=0, sum=0000
    r_a = 4'd5;  r_b = 4'd3;  #10; // carry=0, sum=1000
    r_a = 4'd15; r_b = 4'd1;  #10; // carry=1, sum=0000
    r_a = 4'd15; r_b = 4'd15; #10; // carry=1, sum=1110

    // 뺄셈
    r_add_nsub = 0;
    r_a = 4'd5; r_b = 4'd3; #10; // carry=0, sum=0010
    r_a = 4'd7; r_b = 4'd7; #10; // carry=0, sum=0000
    r_a = 4'd3; r_b = 4'd5; #10; // carry=1, sum=1110
    r_a = 4'd0; r_b = 4'd1; #10; // carry=1, sum=1111

    // 같은 값 덧셈 뺄셈
    r_a = 4'd9;
    r_b = 4'd2;
```

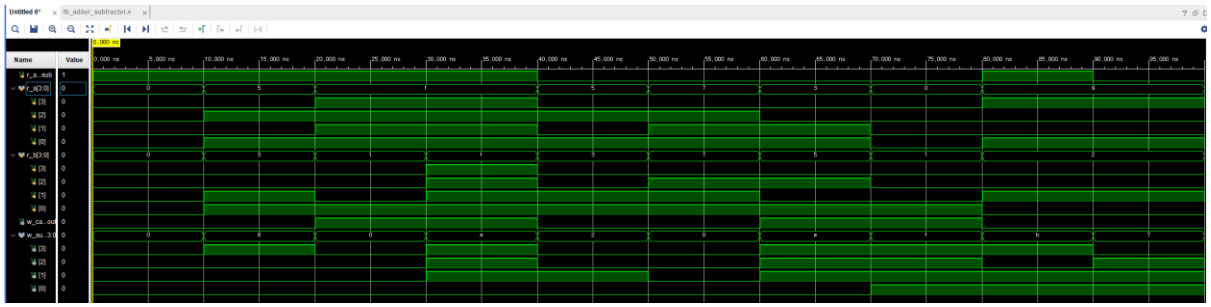
```
    r_add_nsub = 1; #10; // carry=0, sum=1011
    r_add_nsub = 0; #10; // carry=0, sum=0111

    $finish;

    end

endmodule
```

test bench 파형



adder_suber.v

source code

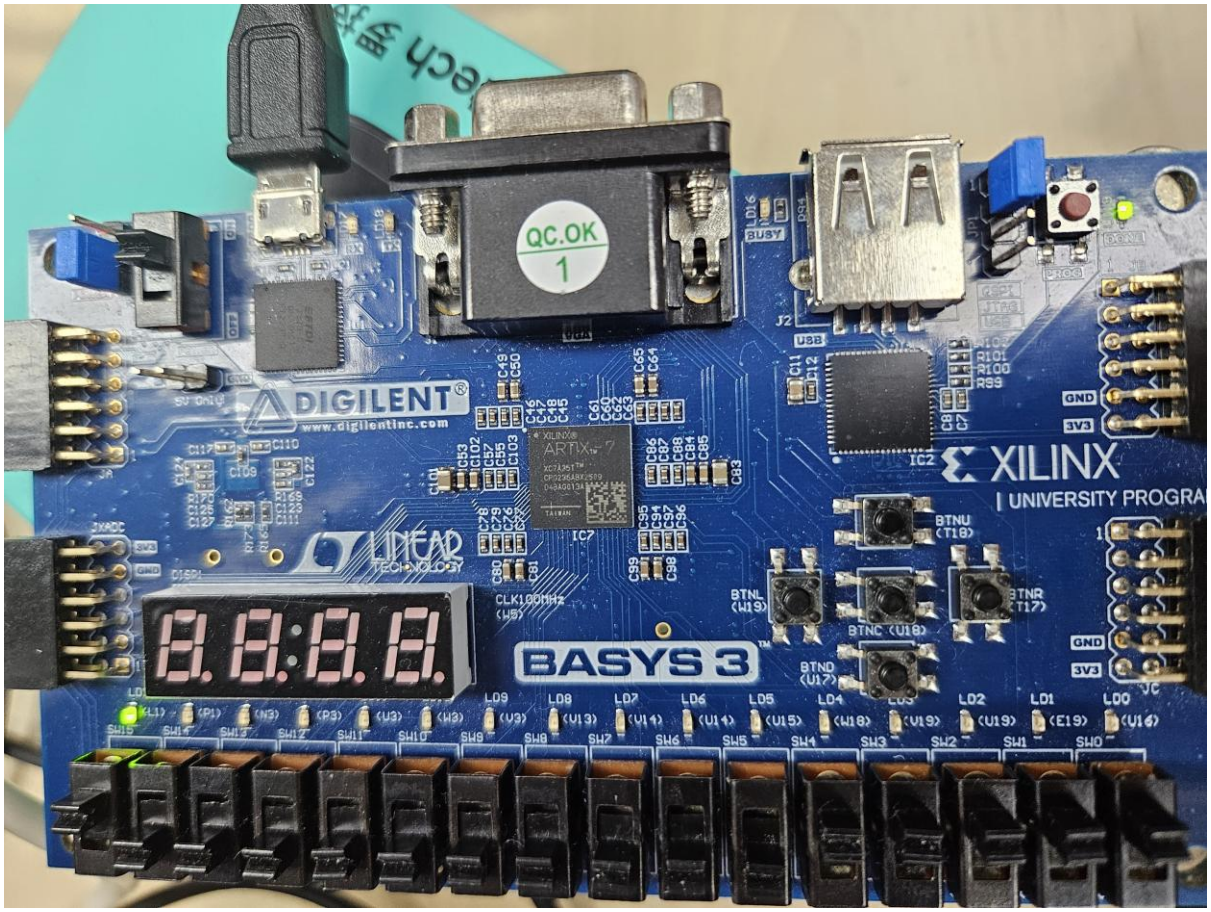
```
`timescale 1ns / 1ps

module adder_subtracter(
    input add_nsub,
    input [3:0] a,
    input [3:0] b,
    output carry_out,
    output [3:0]sum
    );

    reg r_carry_out;
    reg [3:0] r_sum;

    always @(*) begin
        if(add_nsub == 1) {r_carry_out,r_sum} = a[3:0] + b[3:0];
        else {r_carry_out,r_sum} = a[3:0] + ~b[3:0] + 4'b1;
    end

    assign carry_out =  r_carry_out;
    assign sum = r_sum;
endmodule
```



add_subnd = 1 (덧셈)

a = 1111 (15)

b = 0001 (1)

a+b = 16 (10000) -> 자리올림(carry_out 하나만 켜짐)